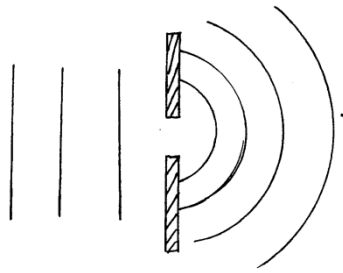


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PHYSICS PAPER 2

MARKING SCHEME

1

2 a) speed = $2d/t$

$$= \frac{400 \times 2}{2.5 \sqrt{}}$$

$$= 320 \text{ m/s} \sqrt{}$$

$$\text{b) } 320 = 2 \frac{(x - 400)}{4.5 \sqrt{}}$$

$$x - 400 = \frac{320 \times 4.5}{2}$$

$$x - 400 = 720$$

$$x = 1120 \text{ m} \sqrt{}$$

3. To concentrate the magnetic field for maximum time linkage

$$4. \quad n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$6/3 \sin 30^\circ = 3/2 \sin r \sqrt{}$$

$$\sin r = \frac{2/3 \times 6/3 \sin 30^\circ}{1}$$

$$= 0.667$$

$$r = 41.813^\circ \sqrt{}$$

5a) A – zinc

B- copper

b) The bulb goes off because of a simple cell polarization and local action

6) In (a) the load is before the fuse so blowing out the fuse does not cut off power from the load . ie the load is still at a higher temperature while in (b) the fuse is before the load so melting of the fuse disconnects the load from the power $\sqrt{}$

2) Pushing the button completes the current making the solenoid to be magnetized .The soft iron armature is attracted towards the solenoid making the hammer to strike the gong .this brakes the contact demagnetizing the solenoid hence the armature is pulled back by the spring . The process repeats itself

8) x-rays $\rightarrow$ uv rays $\rightarrow$ red light $\rightarrow$ micro waves

9.) more information can be transmitted at the same time
O loss of energy during transmission

$$10) \quad RT = 3 + 2 = 5 - 2$$

$$I = \frac{V}{R_T} = \frac{10}{5} = 2 \text{ A} \sqrt{}$$

Pd across 8-2 resistor is zero

$$11. \quad 9a = 234$$

$$a = \underline{234}$$

1

$$\begin{aligned} &9 \\ &= 26 \\ &b+2 = 84 \\ &b=84-2 \\ &= 82 \end{aligned}$$

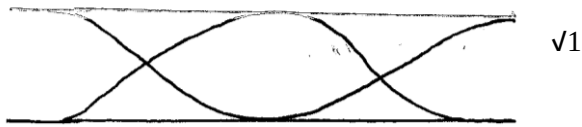
12. They are charged as they can be deflected in both magnetic and electric field
They carry energy

- 13 a) Is provide coherent sources ✓1
ii) - Alternating dark and bright fringes✓1
- Dark fringes due to destructive interference✓1
- Bright fringes due to constructive interference✓1

iii) I. Increased distance between the fringes ✓1
II) Coloured fringes are formed✓1

- b) i) Sound becomes fainter✓1
on cooling ,steam condenses creating vacuum hence cant be transmitted ✓1

ii)



14 a) i) = $\frac{3.5(i)}{4.5}$ ✓1
= 0.778mf ✓1

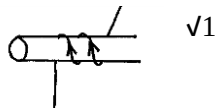
ii) $E = \frac{1}{2}cv^2$ ✓1
= $\frac{1}{2}(3.5/4.5)^{102} \times 10^{-6}$ ✓1
= 3.89×10^{-5} ✓1

b) i) $-v = \frac{1}{\text{slope}}$
= $\frac{12-8}{2.6-4}$ ✓1
= 0.352Ω ✓1

ii) $E = 6.8 \times 10^{-1}$ v ✓1

- 15a) i) Lenz-Ten – the direction of induced current thick that is opposes the causing it ✓1

ii)



- b) i) deflects momentarily✓1
ii)

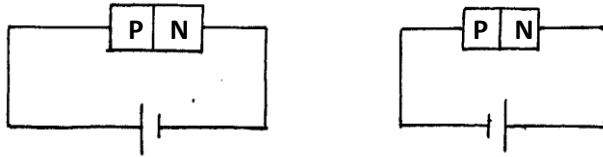
c) i) $\frac{N_s}{N_p} = \frac{V_s}{V_p}$
 $50/N = 6/240$ ✓1

$N = 2000$ turns✓1

ii) $\text{pour/pin} \times 100\%$
 $60 \times \frac{24}{\text{Pin}} \times 100$ ✓1

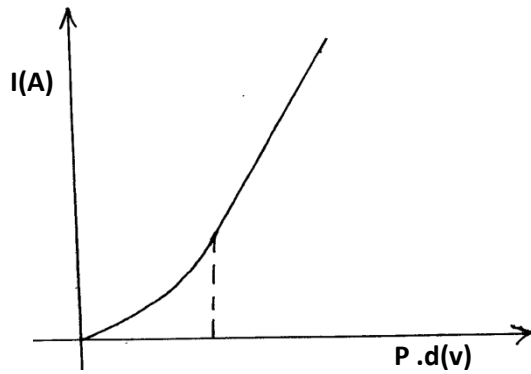
$$= 40 \text{ W} \sqrt{1}$$

16. a

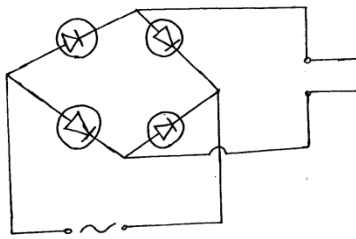


- In forward biasing p- region is connected to positive terminal of external power where n- region is connected to negative terminal to potential barrier reduced
- in reverse bias of potential barrier increased

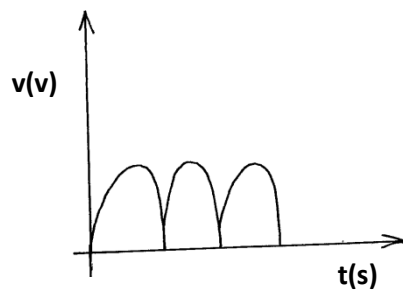
ii)



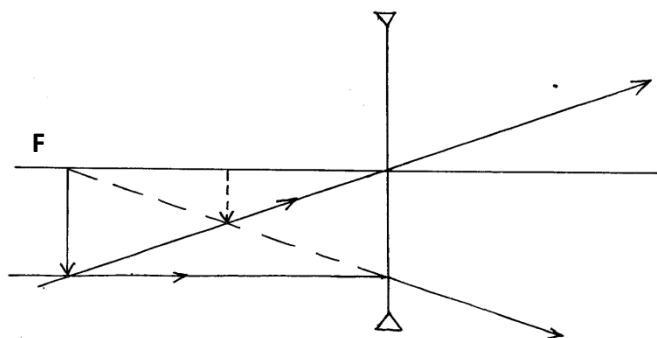
b) i)



ii)



17. a)

b) $u = 12 \text{ cm}$

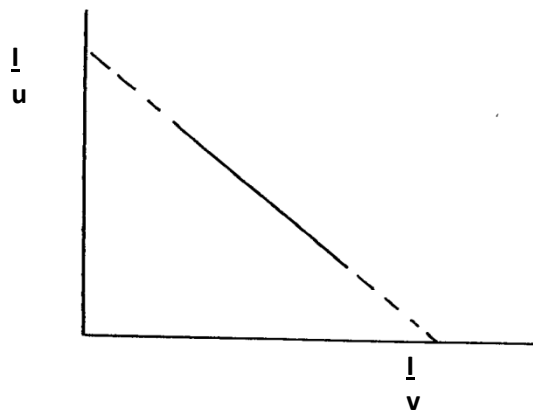


$$v = -36\text{cm}$$

u

$$\begin{aligned} f &= 1/u + 1/v \\ &= 1/2 + 1/-36 \\ &= 1/12 - 1/36 \\ &\quad 3-1/36 \\ &= 2/36 \\ F &= 36/2 \\ &= 18\text{ cm} \end{aligned}$$

c)



$$\begin{aligned} \text{(i)} \quad \frac{1}{u} - \text{intercept} &= 0.07 \\ \frac{1}{f} &= 0.07 \end{aligned}$$

$$\begin{aligned} f &= 0.07 = 1/0.07 \\ &= 14.29\text{cm} \end{aligned}$$

$$\text{(ii)} \quad \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{14.29} = \frac{1}{20} + \frac{1}{v}$$

$$\frac{1}{V} = \frac{1}{14.29} - \frac{1}{20}$$

$$= 0.07 - 0.05$$

$$= 0.02$$

$$\begin{aligned} V &= \frac{1}{0.02} \\ &= 50\text{cm} \end{aligned}$$

$$18 \text{ a) i) } f_0 = 3.5 \times 10^{14} \text{ Hz}$$

$$\text{ii) } h/e = \text{slope}$$

$$h = 6,737 \times 10^{-34} \times 3.5 \times 10^{14}$$

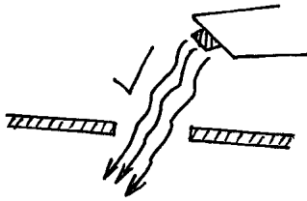
$$= 2.358 \times 10^{-17}$$

$$\text{iii) } W_0 = h f_0$$

$$6.737 \times 10^{-34} \times 3.5 \times 10^{14}$$

$$2.358 \times 10^{-17} \text{ J}$$

b) i)



ii) for x-ray beam to pass through the window ✓1

iii) cools the anode ✓1

iv) prevents collision between electrons and air ✓1

v) force electrons onto the target ✓1